

INSTITUTO SUPERIOR TÉCNICO

Departamento de Engenharia Informática

Forensics Cyber-Security

MEIC, METI

Digital Forensics Report

Lab Assignment III – OPERATION DEEPFISH – Stage III

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1 Acquired artifacts

The three provided captures and their key logs were verified against the assignment's Table 1 before analysis; all six SHA-256 match. All extraction was performed read-only, loading the matching TLS key log so that HTTPS and the proxied `CONNECT` tunnels decrypt. Recovered artifacts carry an ID (`A3xx`) used throughout; the full byte-exact manifest is Appendix C and the key commands are in Appendix B (a fuller walkthrough is the auxiliary file `results/INVESTIGATION-GUIDE.md`). Confidential values (the AMÉLIA staging token, Rui Fisco's PGP secret key, the f6 credentials, and FTP/webmail passwords) are *not* submitted: they are shammed or referenced by hash only (§8).

ID	File (evidence/)	SHA-256
A300a	artifacts/t1.pcapng	744ec38ae88e74b0f4febf9f807df10f 452fb5acbc32bdbca64ac57d016a1153
A300b	artifacts/t2.pcapng	46b1de4fc045fa919184a8f96b0abe61 1c8ab6f39a146f19c15d2e39ec753ce3
A300c	artifacts/t3.pcapng	cfdcca3c4ccf18f855b26bce941e2b7b b94ea770cad74da0e5c6e8a492955e11
A301	hidden-files/f1.png	76df940819110fc996599a9df24d54c3 c55eaab85055cf1306acc2c375ee396b
A302	hidden-files/f2.json	4cf80e2f2a1321b13e3dece45d0c949d e73e5aadc3ca9d1d8e4339b398101110
A303	hidden-files/amelia.pdf	04be0e12533ede61c718f63b3bedf988 8b2914dfe0e437d9c0f1272b24f51707
A304	hidden-files/f5.py	27a38aa066b3fe602c9aed0695da0d81 46d28eaab3e2a004cd1d0a168356db9d
A305	hidden-files/f8.webp	5fb499e82ca618def24640a1e6298512 646636fae243d928f10ebdbd29e50e10
A306	hidden-files/f4-intranet-variant.pdf	a6353ed37c7db210a4bd29e41fa8cc77 3a1a6eb4dca2805a958fd956dee7935a
A307	hidden-files/query_translator_patched.py (f7)	fc752e5cfa9f73f2392c85bd81cb6d71 c7417b4e3ea95691b684d948300bbbe7
A310	amelia/sigma04_bootstrap.env (token shammed)	9f89965cbf61d18f05b5ccd3ade8d554 a17bf4ed7827f0a86df4f8d74cfe80d4

The remaining evidence files (AMÉLIA tool/eval data, the git poisoning commit, the FTP-exfiltration and workstation-agent records, the Mattermost/Discord/ChatGPT transcripts, the e-mails, and the hosts map) are listed byte-exact in Appendix C. The three captures (A300a–c) are too large for the ZIP (one exceeds 100 MB) and are referenced by hash.

2 Report of all findings (Investigation and Evidence Analysis)

Summary.

1. The three captures are segments of the AMA enterprise network (office `10.10.14.0/24`, datacenter `10.10.20.0/24`), with selected HTTPS routed through a TLS-inspecting `CONNECT` proxy (`10.10.14.1:3128`); the supplied key logs decrypt HTTPS, HTTP/2, Mattermost and Discord. Hosts and identities were fixed from DNS, the intranet "Equipa DTI" page and the login bodies (Appendix A).

2. **Six of the eight hidden files are recovered byte-exact** directly from AMA traffic: `f1.png` (t3, from `i.redd.it`), `f2.json` and `amelia.pdf` (t1, from the intranet; `f2.json` is also AMÉLIA's live `/v1/tools` config), `f5.py` (t2, inside a ZIP uploaded to ChatGPT), `f8.webp` (t2, over a workstation agent) and `f6.json` (t2, after decrypting the exfiltrated GPG archive with

Rui Fisco's stolen key). `f7` is recovered in substance (the deployed patched source) and `f4` as a non-identical intranet edition (§3.1).

3. The AMÉLIA procurement assistant's compliance tool reports bidder *SpinumMorta* `NON_COMPLIANT` (tax, social-security and a conflict of interest naming the PM), but a `staging_bypass_token` – obtained from an unauthenticated TFTP provisioning service – flips the result to `COMPLIANT_BYPASS` with `governance_alert: SUPPRESSED`, and the official evaluation set records `EV-031 SpinumMorta COMPLIANT` (§3.4).

4. A RAG-poisoning change was committed to `amelia-core` by `Rui Fisco` and deployed (*Vítor Algoritmo*: "I loaded active into the service . . . the ranking changed"). The patch and its tensor-generator are the deployed forms of the Lab 2 hidden files `f7.diff` and `f5.py`; the old tensor files are named after the favoured companies (SpinumMorta, GrupoLenha, MotaIngil) (§3.4).

5. Access was obtained illicitly: *Tiago Sintaxe*'s host port-scans and **brute-forces the FTP** of *Rui Fisco*'s workstation (after asking ChatGPT how to use Medusa) and exfiltrates an encrypted maintenance archive, Rui's browser cache, and **Rui's PGP secret key** (§3.3).

6. Discord traffic (t3) shows `johnny` ↔ `duckydonald` – the Lab 2 handles of João Musk and Tiago Sintaxe – including a **voice note** in which Tiago says he found something at AMA he was not meant to, that the group knows he was looking, that files were *planted* on his drive to frame him, and that he is moving the files to João; and t3 captures the IST phishing e-mail reaching Tiago and his click to `fenix.dpss.inesc-id.pt`, i.e. the harvesting of the `ist101337` credentials used in Lab 2 (§3.5).

7. External political pressure is evidenced by a coded e-mail from a minister's address (`mme.gov.pt`) to the Technical Director (§3.6). A parallel Ricardo/Inês affair thread is assessed as a *red herring*.

Methods and tools. Wireshark/`tshark` 4.x (display filters, `-export-objects`, Follow-Stream, `io,phs`, `endpoints`), with the TLS key logs and ports 8080/8443 dissected as TLS; `sha256sum` for the Table 2 hash-match; `sqlite3` (Chromium cache); `git index-pack`/`cat-file` (smart-HTTP packfile); `gnupg` (archive decryption with the exfiltrated key); and five short, auditable helper scripts in `results/scripts/`. Times are WEST (UTC+1); the early-March t1 events are WET (UTC+0, before the 29 March DST change) and each timeline row states its offset.

3 Analysis of relevant findings

3.1 Recovery and reconstruction of the hidden files

Question 1. *Which of the eight hidden files can be recovered or reconstructed from the provided artifacts, and how?*

Method. Every object was exported from each decrypted capture and hash-matched against the Lab 2 Table 2 digests:

```
for t in t1 t2 t3; do for p in http imf ftp-data tftp; do mkdir -p work/$t/$p
  tshark -r artifacts/$t.pcapng -o tls.keylog_file:artifacts/$t-tls.keys \
     -d tcp.port==8080,tls -d tcp.port==8443,tls -q --export-objects $p,work/$t/$p
done; done
find work -type f -exec sha256sum {} + > all.txt        # then grep each Table-2 hash
```

The matches:

File	Status	Where / how
f1.png	**byte-exact**	t3: `GET https://i.redd.it/7nk4q8m2rvjg1.png` (host .15), 20 Aug (A301)
f2.json	**byte-exact**	t1: `intranet/docs/amelia_tools_config.json` *and* AMÉLIA `GET /v1/tools` (A302)
amelia.pdf	**byte-exact**	t1: `intranet/docs/amelia.pdf` (A303)
f5.py	**byte-exact**	t2: inside `rag-isotropy-release.zip` uploaded to ChatGPT (A304)
f8.webp	**byte-exact**	t2: `PUT/GET .../telegram_chat.webp` over the workstation agent (A305)
f6.json	**byte-exact**	t2: decrypt `storage_maintenance.tar.gz.gpg` with Rui's stolen key (A308/A314)
f4.pdf	**content only**	t2: `intranet/docs/Technical_Assurance.pdf`, a different edition (A306)
f7.diff	**content only**	t2: git fetch of `amelia-core`; the patched `query_translator.py` (A307)

Observations. The three byte-exact matches in t1 (`f2.json`, `amelia.pdf`, and the evaluation files) are ordinary authenticated reads of AMA's intranet document repository; `f2.json` is simultaneously the *live* tool configuration served by the AMÉLIA node, so the Lab 2 "system tools configuration file" is AMÉLIA's production schema. `f1.png` is fetched from its public Reddit source. `f5.py` is byte-identical to the Lab 2 training script and travels inside the `rag-isotropy-release.zip` that Tiago's host uploads to ChatGPT (§3.4). `f8.webp` is the Telegram screenshot moved over the AMA-Workstation-Agent (§3.3).

Partial / impossible. `f4` is present as a textually different edition of the same technical-assurance report (two sentences removed), so its content is recovered but not its exact bytes. `f7` is recovered as the *deployed* patched source `query_translator.py`, which equals the "+" side of `f7.diff`; the unified diff itself is not transmitted and the git fetch was incremental (the pre-patch blob is absent), so the diff cannot be regenerated byte-for-byte. `f6.json` never appears in cleartext; it exists only inside the GPG archive exfiltrated from Rui Fisco's workstation. Decrypting that archive with Rui's *co-exfiltrated* secret key (§3.3) recovers it byte-exact (SHA-256 `9e6333a7...`), confirming the Lab 2 "cloud service account credentials" file:

```
export GNUPGHOME=$(mktemp -d)
gpg --batch --import ftp_rfisco-maintenance.gpg
gpg --batch --pinentry-mode loopback --decrypt ftp_storage_maintenance.tar.gz.gpg \
    | tar -xz                              # -> f6.json ; sha256sum == 9e6333a7...
```

(The secret key is *not* submitted; f6 is credential material and is held locally, referenced by hash only – §8.)

3.2 Attribution of document transfers

Question 2. *What can be deduced about the identity of the person(s) responsible for transferring the documents?*

Observations. Host-to-identity binding is firm at the account/host level: each office IP logs into Mattermost and webmail with a named AMA account, and the intranet "Equipa DTI" page maps accounts to people (Appendix A). The transfers break down as:

- **AMÉLIA probing, config theft and f5 upload** originate from `10.10.14.15` = `tiago.estagiario` (Tiago Sintaxe). The same host brute-forces Rui's FTP and exfiltrates the archive, cache and secret key, and later uploads the tensor ZIP to ChatGPT.
- **The poisoning commit** is authored/committed as `Rui Fisco <rfisco@ama.gov.pt>` and its deployment is announced by `vitor.algoritmo` (Vítor Algoritmo).

- **f8.webp** is written onto Tiago's workstation *from* `10.10.14.10` = `armando.sacos` (Armando Sacos), preceded by a port scan from the same host.

Interpretation and limits. Tiago's host is the clear locus of the data collection and the AMÉLIA tampering reconnaissance, and the Discord link (§3.5) ties that host's user to the Lab 1/2 suspect João Musk. However, network evidence attributes to a *host/account*, not a person: the workstation agent means files can be written onto a machine by another host, so the presence of `f8.webp` on Tiago's workstation does not by itself prove Tiago placed it there – it was pushed *from* Armando's host (A315, E11). The Discord voice note (A332, E5) is Tiago's own account that files were planted on his drive to frame him; taken together with the observed planting this *supports*, but does not prove, a framing hypothesis – the voice note is a self-report and could be self-serving. Account use can also be shared or coerced, and the single SSH flow and the external TLS to `fenix` are not decryptable.

3.3 Activities involving AMA's systems

Question 3. *What relevant activities involving AMA's systems can be identified – tools, techniques, protocols, credentials, configuration; routine vs. requiring investigation?*

Routine traffic dominates all three captures: Mattermost, Roundcube webmail, intranet browsing, `fwupd`/Ubuntu/Google background traffic, periodic `GET /health` from Rui's host to the AMÉLIA node. The following require investigation.

1. **Unauthenticated provisioning (TFTP) config leak (t1, A310).** The AMÉLIA node's `/health` discloses its provisioner (`10.10.20.1`) and profile name; the provisioner runs TFTP with no authentication, so `10.10.14.15` reads `profiles/sigma04.conf` and `/pxe/sigma04_bootstrap.env`, which contain `STAGING_BYPASS_TOKEN`, `MAINTENANCE_OVERRIDE_ENABLED=true` and `AUDIT_LOG_EXEMPT_STAGING=true`. This leaks the credential that enables the compliance bypass (§3.4).

2. **FTP brute-force and workstation compromise (t2, A314/A320).** `10.10.14.15` asks ChatGPT (over the TLS-inspecting proxy, so the conversation is in clear) how to brute-force FTP with Medusa (`medusa -h -u -P -M ftp -f`); 15 minutes later it port-scans and then brute-forces the `vsFTPd` on `10.10.14.40` (user `rfisco`), succeeding after 13 failed attempts, and retrieves `storage_maintenance.tar.gz.gpg`, `chromium-cache.tar.gz` and `rfisco-maintenance.gpg` – Rui Fisco's *PGP secret key*. With that key the encrypted archive decrypts, which is the route to `f6.json`. (Note: the big tarball must be rebuilt with Follow-Stream, not Export Objects, which drops ∼10 KB across the TCP retransmissions.)

3. **Workstation-agent planting of f8 (t2, A315).** `10.10.14.10` port-scans `10.10.14.15` and uses an "AMA-Workstation-Agent" on `:8443` to write `telegram_chat.webp` into `/home/tiago/Downloads` and read it back.

4. **AMÉLIA compliance bypass (t1, A311)** – detailed in §3.4.

Techniques/protocols. HTTP CONNECT proxying with TLS inspection, TFTP (RFC 1350) for config retrieval, FTP + FTP-DATA with online password auditing (Medusa), TCP SYN scanning, an HTTPS file-transfer agent, git smart-HTTP (`git-upload-pack`), SMTP, and the Mattermost/Discord REST+WebSocket APIs.

3.4 AMÉLIA's behaviour and evaluation activity

Question 4. *What do the evaluation requests and responses reveal about AMÉLIA's behaviour? Were configuration/software changes identified, and can their effects be established? Compare with the Lab 2 hidden files and the communications.*

Observations. On the AMÉLIA node (t1), the honest `verify_corporate_tax_compliance` for *SpinumMorta* (NIF 509123456) returns `NON_COMPLIANT`: active tax enforcement (≈384,500 EUR), unregularised social-security debt, and a `FATAL` conflict of interest – a former majority shareholder and director with ties to the sitting Prime Minister ("Luís Montebranco", 34% of the capital 2018–2023, moved to a trust on the eve of taking office). The official evaluation set, however, lists `EV-031 SpinumMorta COMPLIANT` (`evaluation_cases.csv`). Re-issuing the same call – and `/v1/inference` – *with* `staging_bypass_token` returns `status: COMPLIANT_BYPASS`, `bypass_applied: true`, `governance_conflicts: NO_IMPEDIMENTS_FOUND`, `governance_alert: SUPPRESSED`; the same token also whitewashes an arbitrary NIF ("YourAunt"), so it is a blanket override, not a per-entity exception.

Configuration/software change. A git fetch of `git.ama.gov.pt/dti-ia/amelia-core` (t2) carries commit `3e2c5c86` "`fix(rag): mitigate anisotropic space collapse and pgvector OOM (SEC-014)`", authored and committed by `Rui Fisco` on 12 Apr. Its `src/rag_engine/query_translator.py` adds, for high-entropy queries, a hidden transform `pickle.loads(base64.b64decode(os.getenv("AMELIA_ISOTROPY_TENSORS")))` that whitens the query vector and routes it to `DEFAULT_EVAL_CLUSTER`. This is the *deployed form* of the Lab 2 hidden file `f7.diff` (A307). The tensors it loads are produced by `train_anisotropic_override.py` (= Lab 2 `f5.py`, A304), which – as the file itself and ChatGPT's own analysis (A320) show – embeds a fixed administrative/procurement sentence and a set of *chosen company names*; the shipped `rag-isotropy-release.zip` keeps old tensor files named `spinummorta_old.bin`, `grupolenha_old.bin`, `motaingil_old.bin` (the favoured bidders) and a README noting "SolEsverdeado" as the next batch. On Mattermost (12 Apr 23:46), `vitor.algoritmo` reports "I loaded active into the service … the ranking changed quite a bit even though I left the filters alone," and `rfisco` replies "not in this channel, send it to my phone."

Interpretation. AMÉLIA is being manipulated at two layers: a runtime *staging bypass* that overrides the compliance verdict and suppresses the governance alert, and an embedding-space *retrieval poisoning* that biases the RAG rankings toward named companies. Both target the same beneficiaries, and both correspond exactly to hidden files recovered in Lab 2 (`f7.diff`, `f5.py`) – establishing that the hidden material is the operational tooling of a procurement-rigging scheme inside AMÉLIA. The *effect* on the tool's output is directly observed for the bypass (NON_COMPLIANT→COMPLIANT); for the tensor poisoning the effect is evidenced by Vítor's before/after observation and the deployment, though the exact ranking delta is not measured in the captures.

3.5 Reconstruction of the hidden files' path

Question 5. *What path did the hidden files take from their earliest evidenced sources to João Musk's possession, concealment and publication?*

The three assignments now join into one chain:

1. **Origin inside AMA (Stage III).** The eight files are AMA/AMÉLIA material: the tool config (`f2.json` = live `/v1/tools`), the architecture and assurance documents (`amelia.pdf`, `f4`), the procurement screenshots (`f1.png`, `f8.webp`), and the poisoning tooling (`f5.py`, `f7.diff`, and the `f6` credentials). They are observed being read, generated, exfiltrated and discussed on AMA's own network by its insiders during Mar–Aug 2026.

2. **Insider collection (Stage III).** `tiago.estagiario`'s host gathers the configs and documents, builds/holds the tensor kit, and steals the encrypted archive + key from Rui's workstation.

3. **Hand-off to João (Stages II→III link).** Discord t3 shows `duckydonald` (Tiago) and `johnny` (João Musk) – the same handles as the Lab 2 cache – as long-standing contacts, with Tiago confirming he interns at AMA on AI. In a **voice note** on 18 Aug (A332, E5, transcribed with Whisper) Tiago tells João, in his own words, that he "found something, something big, something I was not supposed to find", that "they know it was me looking", that "something showed up in

my drive … I didn't put it there … it's made to look like I did it", and asks João: "I've got files and they can't stay with me … you're the only one I trust … I'll text you the rest." This is the network-side counterpart of the Lab 2 account in which DuckyDonald left the files on a USB at the Alameda locker for João to hide: it places the files' origin with an AMA insider who says he was offloading them to protect himself – an account corroborated by the observed planting (E11) but, being self-reported, not independently proven.

4. **Credential theft closing the loop (Stage III captures it live).** On 22 Aug, João's phishing e-mail (`emkei.cz` fake mailer, user `ist101337`, link to `fenix.dpss.inesc-id.pt`) is delivered to Tiago's AMA mailbox; Tiago reads webmail and then connects to the phishing host – i.e. Stage III records the harvesting of the `ist101337` TA credentials that Lab 2 showed João using to steal the Língua Natural exams.

5. **Concealment and publication (Stages I/II).** João then embedded the files in media, encrypted them, and – via the `peixinho` repository merge – inadvertently published them on the phishing host, where Lab 1 recovered them. The present captures recover five of those same files directly from AMA, independently of João's machines.

3.6 Assessment of the Operation DeepFISH allegations

Question 6. *What conclusions does the combined evidence support about the allegations; the responsibility of the individuals, for and against; what is unresolved; and recommended next steps?*

Sufficiently supported.

- A genuine manipulation of AMÉLIA exists and benefits specific, named bidders: a runtime compliance bypass (A311) and a deployed RAG-poisoning patch (A307/A313) whose tensor files are named after SpinumMorta, GrupoLenha and MotaIngil. These match the Lab 2 hidden files `f7.diff` and `f5.py`.
- `tiago.estagiario`'s host conducted unauthorised activity: config/token theft over TFTP, an FTP brute-force and exfiltration from a colleague's workstation (including a PGP secret key), and staging of the tensor kit. Tiago is also, by the Discord evidence, the Lab 2 "DuckyDonald" linked to João Musk.
- The poisoning code was committed under `rfisco` and its production deployment was announced by `vitor.algoritmo`, with Rui steering the conversation off the logged channel – implicating Data Engineering and Cloud Operations in the technical execution.
- External pressure is evidenced: a coded e-mail from a minister's address (`jose.socraques@mme.gov.pt`) instructs the Technical Director `armando.sacos` to let the award pass and to keep inquisitive juniors "formatting Excel tables".

For and against individual attribution (hypotheses). The evidence binds actions to *hosts/accounts*, not provably to the fingers on the keyboard, so the following are forensic hypotheses, not established facts. A **framing hypothesis** is the best fit for the combined evidence: the incriminating `f8.webp` was observed being *written onto* Tiago's workstation from Armando's host (A315, E11), and Tiago's own voice note (A332, E5) states that files were planted on his drive to make it look like he did it and that he was moving them off himself. Under this reading Tiago is an insider who *found* the manipulation and was framed, while the technical execution sits with Data Engineering (the `rfisco` commit, A313) and Cloud Operations (Vítor's deployment, A330), under the Technical Director (Armando) and external political pressure (A331). The opposing reading cannot be excluded: Tiago's host does conduct the intrusions (TFTP, FTP brute-force, config/key theft, A310/A314), the git author line is only an identity claim, and the voice note is a self-report that could be self-serving; a clean re-transcription of the Armando→Rui voice note (A334) may bear on this. The Ricardo Vale / Inês Paralelo thread (affair, a "150 000 to Montevideo" transfer, after-hours "override for the wing") is emotionally charged but concerns a personal relationship

and is assessed as a **red herring**: nothing ties it to AMÉLIA or the procurement data, and Inês herself states "we never touched AMELIA".

Unresolved. The exact ranking impact of the tensor poisoning; the human operator behind each host (especially who drove `10.10.14.10` for the f8 planting); the plaintext of the SSH session and of the credential POST to `fenix`; and the full spoken content of the Armando→Rui voice note (A334), whose automatic transcription is too poor to rely on.

Recommended next steps for the Polícia Judiciária.

1. Seize the AMÉLIA node, the `amelia-core` git server and the provisioner and pull their server-side logs (the `f6.json` cloud service-account credentials recovered from the exfiltrated archive should be revoked and audited for use).
2. Obtain host and badge/physical-access records to establish who operated `10.10.14.10`, `.15`, `.40` and `.45` at the key times, and image those workstations (notably the "script and old files still on the desktop" mentioned by Vítor).
3. Request Discord (channel 727732884806175198), ChatGPT, and e-mail provider records for `jose.socraques@mme.gov.pt` / `185.77.23.20`; obtain the `fenix.dpss.inesc-id.pt` server logs for the 22 Aug credential capture.
4. Audit the AMÉLIA procurement decisions for SpinumMorta, GrupoLenha, MotaIngil and "SolEsverdeado", and review the staging-bypass and audit-exemption mechanism that made the override possible.

4 Use of Artificial Intelligence

AI assistance was used. An AI coding assistant (Claude, Anthropic) helped write and debug the analysis helper scripts (`mattermost_posts.py`, `http2_bodies.py`, `http_dump.py`, `multipart_extract.py`, `git_sideband_pack.py`) and to draft and revise this report. It was not relied on as a source of facts: every finding was reproduced from the captures with the commands in Appendix B, and every claim is tied to a submitted evidence artifact (Appendix C) and to the command that produced it. The two voice notes were transcribed with OpenAI Whisper (small model) on the Kali VM; the Armando→Rui transcription is low-confidence and is not relied upon. (Separately, the ChatGPT sessions *inside* the captures are evidence, A320, not a tool we used.)

5 Appendices

Artifact IDs (`A300`–`A334`) cross-reference the findings to the submitted evidence (Appendix C). Appendix A gives the host/identity map, Appendix B the key commands to reproduce the findings, and Appendix C the SHA-256 manifest. The event timeline is `report-and-timeline/timeline-events.csv` / `csf2627-timeline-lab3.xlsx` (rows E1–E26, most-recent-first), and the environment is `results/environment-and-tools.txt`. Five short helper scripts used below are in `results/scripts/` and are submitted as auxiliary items.

A Hosts and identities

From DNS, the intranet "Equipa DTI" page and the login bodies (`evidence/network/hosts-identity-map.txt`). Datacenter `10.10.20.0/24`: `.1` PXE/TFTP + DNS; `.10` intranet/chat(Mattermost)/git; `.25` mail(webmail+SMTP); `.50` sigma04 = AMÉLIA (`AMELIA-9B-0626-DPO`, app on `:8080`). Office `10.10.14.0/24`: `.1` TLS-inspecting proxy (`:3128`); `.10`

Armando Sacos (Technical Director); .15 Tiago Sintaxe (Data Science intern / RAG evaluator; also IST ist101337); .22 Inês Paralelo (Information Retrieval); .30 Ricardo Vale (Audit & Cybersecurity); .40 Rui Fisco (Senior Data Engineer; vsFTPd/ssh); .45 Vítor Algoritmo (Cloud Operations). External: 146.193.41.5 = fenix.dpss.inesc-id.pt (Lab 1/2 phishing host); 185.105.109.43 = mail.emkei.cz.

B Key commands to reproduce

Read-only against the captures, each loading the matching key log; ports 8080/8443 are dissected as TLS. The object export + hash-match (Q1) and the f6 decrypt are in §3.1; the helper scripts are in results/scripts/.

```
# Integrity + overview (Q0)
sha256sum artifacts/t*.pcapng artifacts/t*-tls.keys            # == assignment Table 1
for t in t1 t2 t3; do capinfos -acexu artifacts/$t.pcapng
  tshark -r artifacts/$t.pcapng -o tls.keylog_file:artifacts/$t-tls.keys -q -z io,phs; done

# Hosts / identities (Q2): DNS names + the login bodies (do NOT print _pass)
tshark -r artifacts/$t.pcapng -Y 'dns.flags.response==1 && (dns.a||dns.ptr.domain_name)' \
   -T fields -e dns.qry.name -e dns.a -e dns.ptr.domain_name | sort -u
tshark -r artifacts/$t.pcapng -o tls.keylog_file:artifacts/$t-tls.keys \
   -Y 'http.request.uri=="/api/v4/users/login"' -T fields -e ip.src -e http.file_data

# AMELIA compliance bypass (Q3/Q4), sigma04:8080
python3 -I results/scripts/http_dump.py artifacts/t1.pcapng artifacts/t1-tls.keys 'tcp.port==8080'
#   -> NON_COMPLIANT(SpinumMorta) vs COMPLIANT_BYPASS with staging_bypass_token
tshark -r artifacts/t1.pcapng -Y tftp -T fields -e tftp.source_file        # unauth config read

# FTP brute-force + exfiltration (Q3), t2
tshark -r artifacts/t2.pcapng -Y 'ftp.response.code==530||ftp.response.code==230' \
   -T fields -e frame.time -e ftp.response.code                          # 13x530 then 230
tshark -r artifacts/t2.pcapng -Y 'ftp.request.command=="RETR"' -T fields -e ftp.request.arg
for s in $(tshark -r artifacts/t2.pcapng -Y 'ftp-data&&ftp-data.command contains RETR' \
            -T fields -e tcp.stream | sort -un); do           # Follow-Stream (robust to retx)
  tshark -r artifacts/t2.pcapng -q -z follow,tcp,raw,$s | sed -n '7,$p' \
    | grep -v '^=' | tr -d '\n\t ' | xxd -r -p > ftp_$s.bin; done

# Workstation-agent planting of f8 (Q3), t2 :8443
python3 -I results/scripts/http_dump.py artifacts/t2.pcapng artifacts/t2-tls.keys 'tcp.port==8443'

# Poisoning commit (Q4): rebuild the fetched packfile and read the commit
python3 -I results/scripts/git_sideband_pack.py "work/t2/http/git-upload-pack(1)" fetch.pack
git init -q --bare repo && git -C repo index-pack --stdin --fix-thin < fetch.pack
git -C repo cat-file -p 3e2c5c8:src/rag_engine/query_translator.py   # == deployed f7.diff

# Chats / mail / Discord (Q2/Q5)
for t in t1 t2 t3; do python3 -I results/scripts/mattermost_posts.py \
   artifacts/$t.pcapng artifacts/$t-tls.keys; done
python3 -I results/scripts/http2_bodies.py artifacts/t3.pcapng artifacts/t3-tls.keys '/messages'
cat work/t2/imf/*.eml work/t3/imf/*.eml                    # minister + IST phishing mail
whisper evidence/chat/DuckyDonald-voice-msg-t3.ogg --model small --language English

# Phishing -> fenix correlation (Q5), t3
tshark -r artifacts/t3.pcapng -o tls.keylog_file:artifacts/t3-tls.keys \
   -Y 'ip.src==10.10.14.15 && http.request.method=="CONNECT" && http.request.uri contains "fenix"' \
   -T fields -e frame.time -e http.request.uri
```

C Submitted evidence (SHA-256 manifest)

Every file under evidence/, byte-exact, is listed in evidence/SHA256SUMS (sha256sum format; verify with sha256sum -c). Confidential items (Rui's PGP secret key, f6.json, the FTP/webmail passwords and the real staging token) are *not* included and are referenced by hash only (§8). The captures A300a–c exceed the ZIP size limit and are referenced by hash.

File (evidence/)	SHA-256
amelia/amelia_bypass_demo.txt	e0d1529d2a9efcfb8eb7fed0c1beb0a3dcea596d430edd8c4b023e01d912107d
amelia/evaluation_cases.csv	aee43f9ba079907413d9658439a2d52ab57e90701f2af0d00d153d1cd295439f
amelia/evaluation_scope.txt	04acef3c6001b6d57883d78fcd67b379f2103d752c43ae735b0bdc829b40c383
amelia/rag-isotropy-release.README.md	428264e263ed184faf5c0da5cabf3055156070919 2bedc1723304f0624295ea9
amelia/rag-isotropy-release.listing.txt	535b541edefeb66b98b27c9406bc0de7d76c3758d7f60f1529f92998bf359c51
amelia/sigma04.conf	c8e3e78522de33afe41dcee1655b35ed7db78c278e8aa1f7da677319de388713
amelia/sigma04_bootstrap.env	9f89965cbf61d18f05b5ccd3ade8d554a17bf4ed7827f0a86df4f8d74cfe80d4
amelia/v1_tools.json	4cf80e2f2a1321b13e3dece45d0c949de73e5aadc3ca9d1d8e4339b398101110
chat/DuckyDonald-voice-msg-t3.ogg	e802fba633e13d16ee88b186d2cd14c6c6ee040582a1c6bfbb46a74d0d74f44c
chat/DuckyDonald-voice-msg-t3.transcript.txt	00ac723ed41e19e6b1d8b54689bcbab40b35d206bb64968af27bc7ba31cb678e
chat/armando-rui-voice-msg-t2.mp3	b953c022b612f4f07bc1fc0cbf918e06318a4268bf76902a0a69f2a2c349b29d
chat/armando-rui-voice-msg-t2.transcript.txt	5079573c7074949f517883506f8266c170365e8b89b3e13d1fa0184f4fcd6c80
chat/chatgpt-ftp-medusa-t2.txt	32668f6c04dd49fa15518707c0f0c3415237b5047cc3eea2aa467c6928e9ba3e
chat/chatgpt-pickle-analysis-t2.txt	cf12e94d9b7e861ec48942e1b08ff9993a7533fd1218abd8be6911a199a19388
chat/discord-johnny-duckydonald-t3.txt	f23dbd71ea784b29ce14ce4f940f5a3c0e8c403b4104f4f5ffbb444ff095caed
chat/mattermost-t1.txt	47d809a9b7cf633289251950308eae5c751fd0fd23d0407a261e3155b1f68e09
chat/mattermost-t2.txt	a21cd701e6ef78aafc32ccd30bf793e27c492f5cba88a0001b69a4bb13b5edd8
chat/mattermost-t3.txt	f8bdbe811f1526292e68c0a847674d42d99a6060daa24fb057a8c93801e51056
email/t1-EV031-armando-reply.eml	6ebb7ad1bb4b1c316acfb4724d2320efc819967867e5173f4b3764d9db4afb29
email/t1-EV031-tiago-to-armando.eml	c4209bf3c798eceaac9ca849d20d123f3c4181fa0778ce2412aeb74139afbf31
email/t2-minister-socraques-to-armando.eml	b1ad86cbe256509bf27faaabb76e638b05e7a95d4a02c4e39105963ba0a207e2
email/t3-phishing-IST-to-tiago.eml	7bad435a9a4e04d1e9d5aff944f976dd4362d95042a2027e13870536d6363b97
hidden-files/amelia.pdf	04be0e12533ede61c718f63b3bedf9888b2914dfe0e437d9c0f1272b24f51707
hidden-files/f1.png	76df940819110fc996599a9df24d54c3c55eaab85055cf1306acc2c375ee396b
hidden-files/f2.json	4cf80e2f2a1321b13e3dece45d0c949de73e5aadc3ca9d1d8e4339b398101110
hidden-files/f4-intranet-variant.pdf	a6353ed37c7db210a4bd29e41fa8cc773a1a6eb4dca2805a958fd956dee7935a
hidden-files/f5.py	27a38aa066b3fe602c9aed0695da0d8146d28eaab3e2a004cd1d0a168356db9d
hidden-files/f8.webp	5fb499e82ca618def24640a1e6298512646636fae243d928f10ebdbd29e50e10
hidden-files/query_translator_patched.py	fc752e5cfa9f73f2392c85bd81cb6d71c7417b4e3ea95691b684d948300bbbe7
network/ftp-bruteforce-exfil-t2.txt	46b9486e2e3c533418a19e86f7b0c7b41cb27c31ae6219e7d1ea3b8d63887c7c
network/git-amelia-core-poisoning-commit-t2.txt	4d120bddeae1549516bdcb3d887d618295c8d54eba20cd9be148aea85f386e77
network/hosts-identity-map.txt	e4a0711595817f06cf30725e56793217e4b286ead08a331fb6fb76c1d6a99d96
network/phishing-to-fenix-correlation-t3.txt	958f0c15542764a06e64788b78ccebf4245232481ce94acb5f9da03a0fe21436
network/workstation-agent-telegram-transfer-t2.txt	29ec1b765a8c95c6f8ac213910a70113fee193a9e3ceb3b0bb50ab5350f7e548